

信号与系统

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and Information Engineering

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State variable analysis of the system

- **Basic concepts and general forms of system state variable analysis**
- Establishment of state equation and output equation for continuous time system
- Establishment of state equations and output equations for discrete-time systems
- The solution of state equation and output equation of continuous time system
- The solution of state equation and output equation of discrete-time system

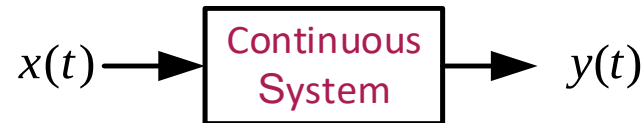


Basic concepts of system state variable analysis

Input/output method:

Features: Single input single output (SISO), only concerned with input and output.

Description: Differential equation (continuous system) or difference equation (discrete system)



$$y^{(n)}(t) + a_{n-1}y^{(n-1)}(t) + \cdots + a_1y'(t) + a_0y(t) \\ = b_mx^{(m)}(t) + b_{m-1}x^{(m-1)}(t) + \cdots + b_1x'(t) + b_0x(t)$$

Where a_i , b_j is constant



$$\sum_{i=0}^n a_i y[k-i] = \sum_{j=0}^m b_j x[k-j] \quad \text{Where } a_i, b_j \text{ is constant}$$



Basic concepts of system state variable analysis

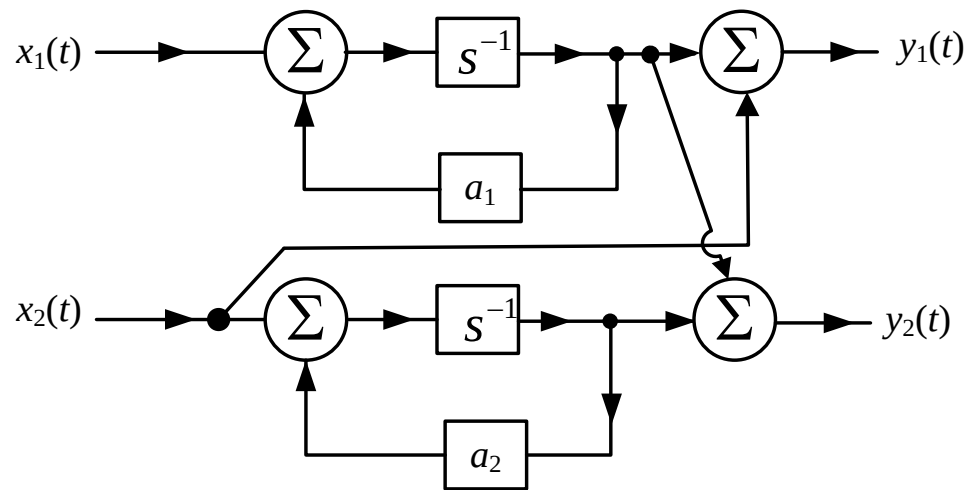
For multiple-input multiple-output (MIMO) systems, the input-output method is difficult to describe.





Basic concepts of system state variable analysis

[Example] The following multiple-input multiple-output system is analyzed

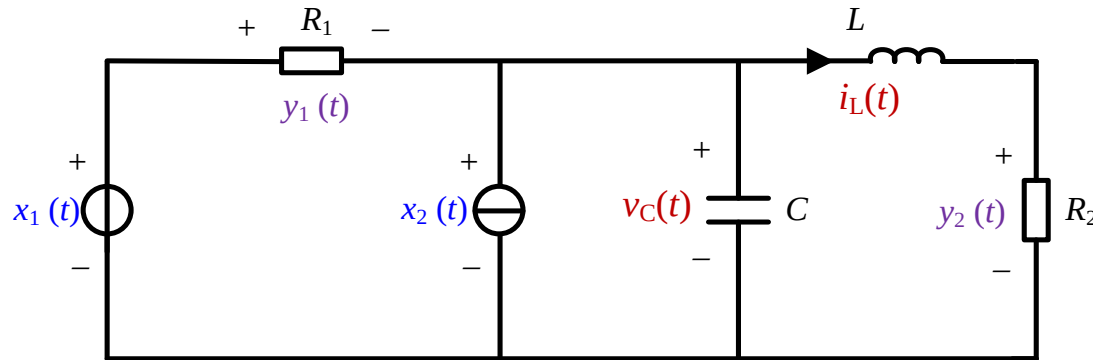


The system has two inputs $x_1(t)$ and $x_2(t)$ and two outputs $y_1(t)$ and $y_2(t)$.



Basic concepts of system state variable analysis

[Example] The following RLC circuit is analyzed



The circuit has two voltage inputs $x_1(t)$ and $x_2(t)$, and two outputs $y_1(t)$ and $y_2(t)$.

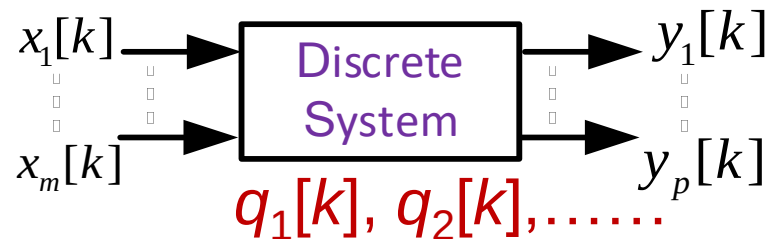
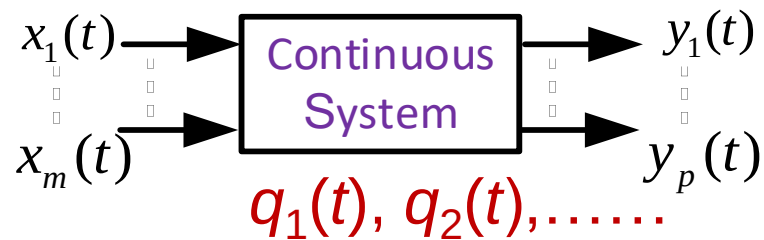
Outputs $y_1(t)$ and $y_2(t)$ can be represented by internal $i_L(t)$, $v_C(t)$, and $x_1(t)$ and $x_2(t)$.



Basic concepts of system state variable analysis

State variable analysis method:

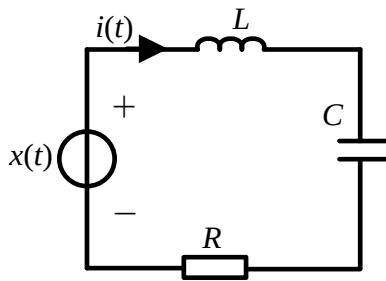
The system is described and analyzed by its internal state variables.



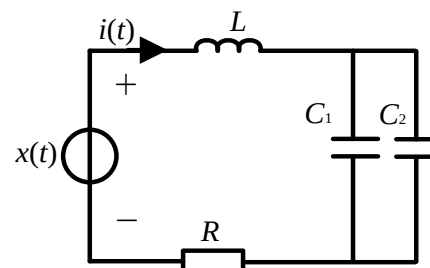


Basic concepts of system state variable analysis

- **State variable $q_i(t)$:** A set of independent (minimal) variables that describe the internal state of a system



图(a)



图(b)

Figure (a) The circuit has two independent energy storage components L and C , so two state variables are required; Figure (b) and Figure (a) are **equivalent circuits**, so Figure (b) also requires only 2 independent variables.



Basic concepts of system state variable analysis

- **State variable $q_i(t)$:** A set of independent (minimal) variables that describe the internal state of a system
- **State vector $q(t)=[q_1(t), \dots, q_n(t)]^T$:** A vector consisting of a set of state variables, a continuous time system described by a differential equation of order n , with n independent state variables.
- **Input vector $x(t)=[x_1(t), \dots, x_m(t)]^T$**
- **Output vector $y(t)=[y_1(t), \dots, y_p(t)]^T$**



Basic concepts of system state variable analysis

Equation of state:

Differential or difference equation that describes the relationship between the state variables of the system and the input variables

First order differential equations (continuous time systems)

First order difference equations (discrete time systems)

Output equation:

An algebraic equation that describes the relationship between the output of a system and the state variables and inputs



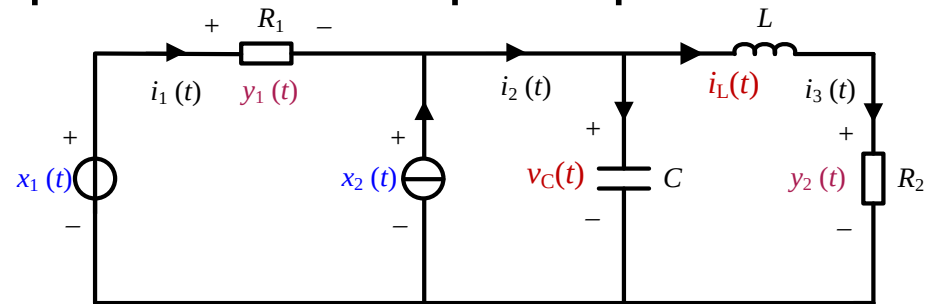
General form of equations of state and output equations

- ✘ General forms of state equations and output equations for continuous time systems
- ✘ General forms of state equations and output equations for discrete time systems



General form of equations of state and output equations

[Example] Write the state equation and output equation of the RLC circuit as follows



Solution: (1) The voltage $v_C(t)$ of C and the current $i_L(t)$ of L are selected as state variables $q_1(t)$ and $q_2(t)$

$$q_1(t) = v_C(t) \quad q_2(t) = i_L(t) = i_3(t)$$

(2) The loop voltage equation is listed $L\dot{q}_2(t) + y_2(t) = q_1(t)$

$$y_1(t) = x_1(t) - q_1(t) \quad y_2(t) = R_2 i_3(t)$$

(3) Column node current equation:

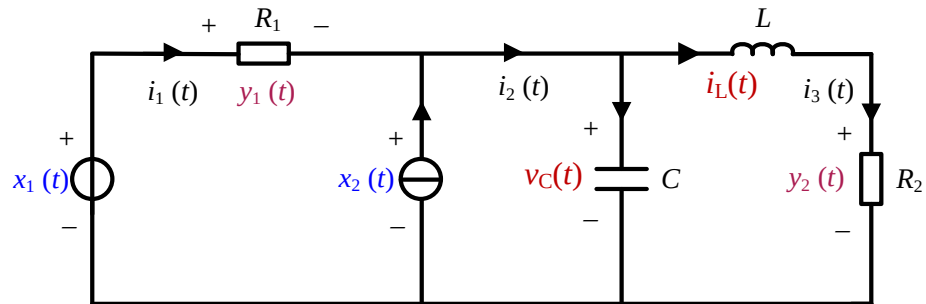
$$i_2(t) = i_1(t) + x_2(t)$$

$$i_2(t) = C\dot{q}_1(t) + i_L(t)$$



General form of equations of state and output equations

[Example] Write the state equation and output equation of the following circuit



After sorting:

$$\dot{q}_1(t) = -\frac{q_1(t)}{R_1 C} - \frac{q_2(t)}{C} + \frac{x_1(t)}{R_1 C} + \frac{x_2(t)}{C}$$

$$\dot{q}_2(t) = \frac{q_1(t)}{L} - \frac{R_2 q_2(t)}{L}$$

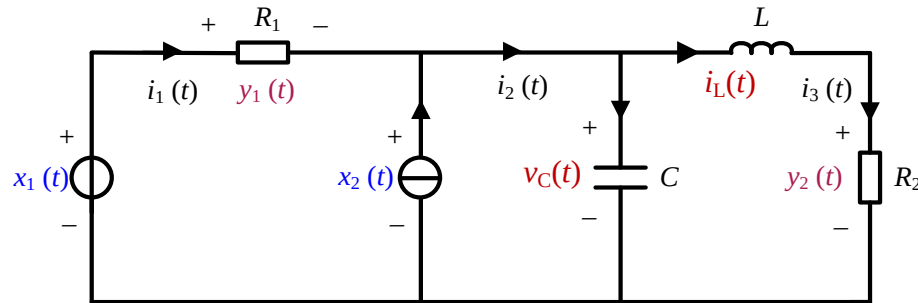
$$y_1(t) = x_1(t) - q_1(t)$$

$$y_2(t) = R_2 q_2(t)$$



General form of equations of state and output equations

[example] Write the state equation and output equation of the following circuit



Written in matrix form:

State equation

$$\begin{bmatrix} \dot{q}_1(t) \\ \dot{q}_2(t) \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1 C} & -\frac{1}{C} \\ \frac{1}{L} & -\frac{R_2}{L} \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} \frac{1}{R_1 C} & \frac{1}{C} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$

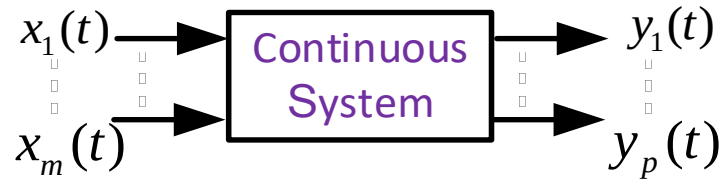
Output equation

$$\begin{bmatrix} y_1(t) \\ y_2(t) \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & R_2 \end{bmatrix} \begin{bmatrix} q_1(t) \\ q_2(t) \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$$



General form of equations of state and output equations

For continuous time systems (m inputs, p outputs)



The state equation of a continuous time system can be expressed by a system of first-order differential equations (n state variables) :

$$\dot{q}_1(t) = a_{11}q_1(t) + a_{12}q_2(t) + \cdots + a_{1n}q_n(t) + b_{11}x_1(t) + b_{12}x_2(t) + \cdots + b_{1m}x_m(t)$$

$$\dot{q}_2(t) = a_{21}q_1(t) + a_{22}q_2(t) + \cdots + a_{2n}q_n(t) + b_{21}x_1(t) + b_{22}x_2(t) + \cdots + b_{2m}x_m(t)$$

$$\vdots$$

$$\dot{q}_n(t) = a_{n1}q_1(t) + a_{n2}q_2(t) + \cdots + a_{nn}q_n(t) + b_{n1}x_1(t) + b_{n2}x_2(t) + \cdots + b_{nm}x_m(t)$$



General form of equations of state and output equations

Output equation of continuous time system:

$$y_1(t) = c_{11}q_1(t) + c_{12}q_2(t) + \cdots + c_{1n}q_n(t) + d_{11}x_1(t) + d_{12}x_2(t) + \cdots + d_{1m}x_m(t)$$

$$y_2(t) = c_{21}q_1(t) + c_{22}q_2(t) + \cdots + c_{2n}q_n(t) + d_{21}x_1(t) + d_{22}x_2(t) + \cdots + d_{2m}x_m(t)$$

$$\vdots$$

$$y_p(t) = c_{p1}q_1(t) + c_{p2}q_2(t) + \cdots + c_{pn}q_n(t) + d_{p1}x_1(t) + d_{p2}x_2(t) + \cdots + d_{pm}x_m(t)$$



General form of equations of state and output equations

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{x}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{x}(t)$$

Matrix form of the equation of state for continuous time systems:

$$\begin{array}{c} \mathbf{A} \\ \left[\begin{array}{c} \dot{q}_1(t) \\ \dot{q}_2(t) \\ \vdots \\ \dot{q}_n(t) \end{array} \right] = \left[\begin{array}{cccc} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{array} \right] \left[\begin{array}{c} q_1(t) \\ q_2(t) \\ \vdots \\ q_n(t) \end{array} \right] + \left[\begin{array}{cccc} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \cdots & \cdots & \cdots & \cdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \end{array} \right] \left[\begin{array}{c} x_1(t) \\ x_2(t) \\ \vdots \\ x_m(t) \end{array} \right] \\ \begin{array}{ccccc} (n*1) & (n*n) & (n*1) & (n*m) & (m*1) \end{array} \end{array}$$



General form of equations of state and output equations

$$\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{x}(t)$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{x}(t)$$

The matrix form of the output equation of a continuous time system:

$$\begin{array}{c} \mathbf{C} \\ \left[\begin{array}{c} y_1(t) \\ y_2(t) \\ \vdots \\ y_p(t) \end{array} \right] = \left[\begin{array}{cccc} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ c_{p1} & c_{p2} & \cdots & c_{pn} \end{array} \right] \left[\begin{array}{c} q_1(t) \\ q_2(t) \\ \vdots \\ q_n(t) \end{array} \right] + \left[\begin{array}{cccc} d_{11} & d_{12} & \cdots & d_{1m} \\ d_{21} & d_{22} & \cdots & d_{2m} \\ \cdots & \cdots & \cdots & \cdots \\ d_{p1} & d_{p2} & \cdots & d_{pm} \end{array} \right] \left[\begin{array}{c} x_1(t) \\ x_2(t) \\ \vdots \\ x_m(t) \end{array} \right] \\ \begin{array}{ccccc} (p*1) & (p*n) & (n*1) & (p*m) & (m*1) \end{array} \end{array}$$



General form of equations of state and output equations

State equation and output equation of continuous time system

State equation: $\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{x}(t)$

Output equation: $\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{x}(t)$

Among:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \cdots & \cdots & \cdots & \cdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \end{bmatrix}$$
$$\mathbf{C} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ c_{p1} & c_{p2} & \cdots & c_{pn} \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1m} \\ d_{21} & d_{22} & \cdots & d_{2m} \\ \cdots & \cdots & \cdots & \cdots \\ d_{p1} & d_{p2} & \cdots & d_{pm} \end{bmatrix}$$



General form of equations of state and output equations

For discrete time systems (m inputs, p outputs)



The state equation of discrete-time system can be expressed by a system of first-order difference equations (n state variables).

$$q_1[k+1] = c_{11}q_1[k] + c_{12}q_2[k] + \cdots + c_{1n}q_n[k] + d_{11}x_1[k] + d_{12}x_2[k] + \cdots + d_{1m}x_m[k]$$

$$q_2[k+1] = c_{21}q_1[k] + c_{22}q_2[k] + \cdots + c_{2n}q_n[k] + d_{21}x_1[k] + d_{22}x_2[k] + \cdots + d_{2m}x_m[k]$$

$$\vdots$$

$$q_n[k+1] = c_{n1}q_1[k] + c_{n2}q_2[k] + \cdots + c_{nn}q_n[k] + d_{n1}x_1[k] + d_{n2}x_2[k] + \cdots + d_{nm}x_m[k]$$



General form of equations of state and output equations

Output equation of discrete-time system:

$$y_1[k] = c_{11}q_1[k] + c_{12}q_2[k] + \cdots + c_{1n}q_n[k] + d_{11}x_1[k] + d_{12}x_2[k] + \cdots + d_{1m}x_m[k]$$

$$y_2[k] = c_{21}q_1[k] + c_{22}q_2[k] + \cdots + c_{2n}q_n[k] + d_{21}x_1[k] + d_{22}x_2[k] + \cdots + d_{2m}x_m[k]$$

$\vdots$

$$y_p[k] = c_{p1}q_1[k] + c_{p2}q_2[k] + \cdots + c_{pn}q_n[k] + d_{p1}x_1[k] + d_{p2}x_2[k] + \cdots + d_{pm}x_m[k]$$



General form of equations of state and output equations

$$\mathbf{q}[k+1] = \mathbf{A}\mathbf{q}[k] + \mathbf{B}\mathbf{x}[k]$$

$$\mathbf{y}[k] = \mathbf{C}\mathbf{q}[k] + \mathbf{D}\mathbf{x}[k]$$

Matrix form of state equation for discrete-time system:

$$\begin{array}{c} \begin{bmatrix} q_1[k+1] \\ q_2[k+1] \\ \vdots \\ q_n[k+1] \end{bmatrix} \\ \begin{matrix} (n*1) \\ (m*1) \end{matrix} \end{array} = \begin{array}{c} \mathbf{A} \\ \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \\ \begin{matrix} (n*n) \end{matrix} \end{array} \begin{array}{c} \begin{bmatrix} q_1[k] \\ q_2[k] \\ \vdots \\ q_n[k] \end{bmatrix} \\ (n*1) \end{array} + \begin{array}{c} \mathbf{B} \\ \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \cdots & \cdots & \cdots & \cdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \end{bmatrix} \\ \begin{matrix} (n*m) \end{matrix} \end{array} \begin{array}{c} \begin{bmatrix} x_1[k] \\ x_2[k] \\ \vdots \\ x_m[k] \end{bmatrix} \end{array}$$



General form of equations of state and output equations

$$\mathbf{q}[k+1] = \mathbf{A}\mathbf{q}[k] + \mathbf{B}\mathbf{x}[k]$$

$$\mathbf{y}[k] = \mathbf{C}\mathbf{q}[k] + \mathbf{D}\mathbf{x}[k]$$

Matrix form of output equation of discrete-time system:

$$\begin{array}{c} \mathbf{C} \qquad \qquad \qquad \mathbf{D} \\ \left[\begin{array}{c} y_1[k] \\ y_2[k] \\ \vdots \\ y_p[k] \end{array} \right] = \left[\begin{array}{cccc} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ c_{p1} & c_{p2} & \cdots & c_{pn} \end{array} \right] \left[\begin{array}{c} q_1[k] \\ q_2[k] \\ \vdots \\ q_n[k] \end{array} \right] + \left[\begin{array}{cccc} d_{11} & d_{12} & \cdots & d_{1m} \\ d_{21} & d_{22} & \cdots & d_{2m} \\ \cdots & \cdots & \cdots & \cdots \\ d_{p1} & d_{p2} & \cdots & d_{pm} \end{array} \right] \left[\begin{array}{c} x_1[k] \\ x_2[k] \\ \vdots \\ x_m[k] \end{array} \right] \\ \begin{array}{c} (p*1) \\ (m*1) \end{array} \qquad \begin{array}{c} (p*n) \\ \end{array} \qquad \begin{array}{c} (n*1) \\ \end{array} \qquad \begin{array}{c} (p*m) \\ \end{array} \end{array}$$



General form of equations of state and output equations

State equation and output equation of discrete-time system

State equation: $\mathbf{q}[k+1] = \mathbf{A}\mathbf{q}[k] + \mathbf{B}\mathbf{x}[k]$

Output equation: $\mathbf{y}[k] = \mathbf{C}\mathbf{q}[k] + \mathbf{D}\mathbf{x}[k]$

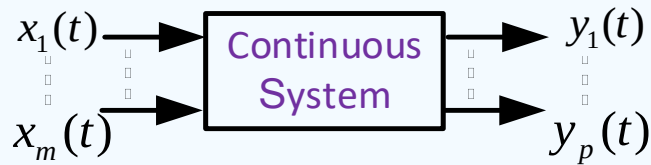
Among:

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \cdots & \cdots & \cdots & \cdots \\ b_{n1} & b_{n2} & \cdots & b_{nm} \end{bmatrix}$$

$$\mathbf{C} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ c_{p1} & c_{p2} & \cdots & c_{pn} \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1m} \\ d_{21} & d_{22} & \cdots & d_{2m} \\ \cdots & \cdots & \cdots & \cdots \\ d_{p1} & d_{p2} & \cdots & d_{pm} \end{bmatrix}$$



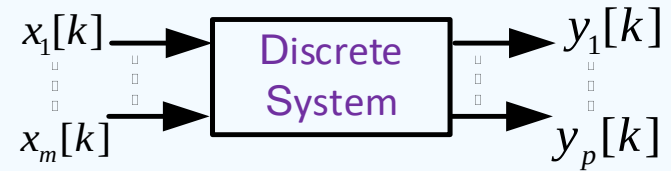
General form of equations of state and output equations



n state variables $q_i(t)$

State equation: $\dot{\mathbf{q}}(t) = \mathbf{A}\mathbf{q}(t) + \mathbf{B}\mathbf{x}(t)$

Output equation: $\mathbf{y}(t) = \mathbf{C}\mathbf{q}(t) + \mathbf{D}\mathbf{x}(t)$



n state variables $q_i[k]$

State equation: $\mathbf{q}[k+1] = \mathbf{A}\mathbf{q}[k] + \mathbf{B}\mathbf{x}[k]$

Output equation: $\mathbf{y}[k] = \mathbf{C}\mathbf{q}[k] + \mathbf{D}\mathbf{x}[k]$



Thanks

Some of the materials quoted in this course are accumulated by the main lecturer over the years, from a variety of media and colleagues, peers and friends of the exchange, it is difficult to attribute them all, so I would like to state and thank you!